CHAPTER

41

Snayugata Vata

Learning Objectives

- To enlist the types of Snayugata Vata
- To know the definition of Akshepaka
- To enlist the types of Akshepaka, Samprapti of Akshepaka
- To know about Apatanaka, types of Apatanaka
- To know about Dandapatanaka, Antarayama, Bahirayama
- To know about Ardita, Samprapti of Ardita, Lakshana of Ardita, Sadhyasadhyata and difference between Lakshana of Ardita as per Sushruta and Charaka
- To know the definition of Pakshaghata, Samprapti of Pakshaghata and Lakshana
- To know about Kampavata
- To know about Gridhrasi and Vishvachi and Lakshana
- To know about differential diagnosis of Gridhrasi and Vishvachi
- To know about Pangutva and difference between Khanja and Pangu

Introduction

Types of Snayugata Vata[1]:

बाह्याभ्यन्तरमायामं खल्लिं कुब्जत्वमेव च।
सर्वाङ्गैकाङ्गरोगांश्च कुर्यात् स्नायुगतोऽनिलः॥ Ca.Ci. – 28/36

As per *Caraka Samhita*, if the vitiated *Vata* afflicts *Snayu*, that will lead to the manifestation of *Bahyayama*[2] (backward bending of the body like a bow), *Abhyantara Ayama* (forward bending of the body like a bow), *Khalli*[3] (muscle cramps in the upper or/ and lower limbs), *Kubjatva*[4] (dwarfness), *Sarvanga Roga*[5] (affliction of whole body by vitiated *Vata Dosha*) and *Ekanga Roga*[6] (affliction of one part of the body by vitiated *Vata Dosha*).

स्नायौ सिरीकण्डराभ्यो दुष्टाः क्लिश्नन्ति मानवम्।
स्तम्भसंकोचखल्लीभिर्ग्रन्थिस्फुरणसुप्तिभिः॥21॥ Ca.Su. – 28/21

GK

1 The term Snayugata Vata refers to the pathological events taking place in the human body. These are actually not the final diagnosis.

2 The features mentioned in Bahirayama and Abhyantara Ayama mimic the features of decorticate and decebrate lesion. The term decorticate lesion refers to the detachment of cerebral cortex from rest of the nervous system and decerebrate lesion refers to the detachment of cerebral hemisphere from the rest of the nervous system. This is indicative of severity of decerebrate lesion over decorticate lesion.

3 In this spectrum of the disease Khalli, the person may have deficiency of essential nutrients such as sodium, potassium and calcium ions which are essential factors for the normal functioning of skeletal, smooth and cardiac muscles and neuronal function. Due to the absence of these essentials, the person may have muscle cramps and poor functioning of the neurons. When compared to the whole body, upper and lower limbs will have maximal functions, in the absence of these electrolytes, the person naturally develops cramps in the upper and lower limbs.

4 In this disease, the person may have short stature i.e. measurement from head to toe will be decreased when compared to the individual of the same age group. When we search the textbooks of physiology, this type of clinical presentation can be seen in growth hormone deficiency in children as well as in adults.

5 This type of clinical presentation can be observed in case of generalized nutritional deficiency either macronutrients or micronutrients or in case of intrinsic aetiology.

6 This type of clinical presentation can be observed in case of localized or extrinsic cause such as trauma to one particular part of the body.

When the vitiated *Vata Dosha* afflicts *Snayu*, *Sira* and *Kandara*, the person will have the manifestation of *Stambha* (stiffness of one part or whole of the body), *Sankocha* (contractures), *Khalli* (muscle cramps in the upper and lower limbs), *Granthi* (nodular swelling), *Sphurana* (abnormal pulsations in unusual parts of the body) and *Supti* (numbness of the one part or whole body).

Akshepaka

मुहुराक्षिपति क्रुद्धो गात्राण्या क्षेपको ऽनिलः॥50॥
पाणिपादं च संशोष्य सिराः सनायुकण्डराः।

Ca.Ci. – 28/50,51

- Vitiated *Vata Dosha* causes frequent involuntary movements in the one part or whole body with symptom free period in between the episodes. This will lead to the dryness of *Pani* (palms) and *Pada* (soles) along with dryness of *Sira*, *Snayu* and *Kandara*.
- The whole clinical presentation is termed as *Akshepaka*[7].

पाणिपादशिरःपृष्ठश्रोणीः स्तनाति मारुतः॥
दण्डवत्स्तब्धगात्रस्य दण्डकः सोऽनुपक्रमः।

Ca.Ci. – 28/51, 52

- When vitiated *Vata Dosha* is responsible for the stiffness of *Pani, Pada, Shira, Prishtha, Shroni*, that will lead the body to become stiff like a stick, this is called as *Dandaka*[8]. This is considered as *Asadhya* (incurable) in nature.

यदा तु धमनीः सर्वाः कुपितोऽभ्येति मारुतः॥50॥
तदाक्षिपत्याशु मुहुर्मुहुर्देहं मुहुश्चरः॥
मुहुर्मुहुस्तदाक्षेपादाक्षेपंक इति स्मृतः॥51॥

Su.Ni. – 1/50,51; Ma.Ni. – 22/27,28

- When vitiated *Vata Dosha* permeates all *Dhamani*, it causes frequent movements, repeated convulsive episodes in the body. Because of producing convulsions, it is known as *Akshepaka*.

वायुरूर्ध्वं व्रजेत् स्थानात् कुपितो हृदयं शिरः॥
शङ्खौ च पीडयत्यङ्गान्याक्षिपेन्नमयेच्च सः॥64॥
निमीलिताक्षो निश्चेष्टः स्तब्धाक्षो वापि कूजति॥
निरुच्छासोऽथ वा कृच्छावुच्छस्यात् नष्टचेतनः॥ 65॥
स्वस्थः स्याद्धृदये मुक्ते आवृते तु प्रमुह्यति॥
कफान्वितेन वातेन ज्ञेय एषोऽपतन्त्रकैः॥66॥

Su.Ni. – 1/64-66

- Aggravated *Vata Dosha* moves from its place upwards to the heart, head and temporal region and afflicts the limbs with convulsions (involuntary movements) and bending of the body.
- The patient will have closed eyes, loses movement or with fixed gaze., makes indistinct sound like pigeon with stooped inspiration or inspires with difficulty while losing consciousness.

कफपित्तान्वितो वायुर्वायुरेव च केवलः॥
कुर्यादाक्षेपकं त्वन्यं चतुर्थममिघातजम्।

Ma.Ni. – 22/37

Apatanaka

दृष्टि संस्तभ्य संज्ञां च हत्वा कण्ठेन कूजति।
हृदि मुक्ते नरः स्वास्थ्यं याति मोहं वृते पुनः॥31॥
वायुना दारुणं प्राहुरेके तदक्षतानकम्।

Ma.Ni. – 22/31-32

- The person will have steadiness of the gaze without proper knowledge out of the fixed gaze; sounds will be produced through the throat (e.g. stridor); altered state of consciousness
- This disease is manifested by abnormally vitiated *Vata Dosha* and it is a serious disease.

Sadhyasadhyata of Apatanaka

गर्भपातनिमित्तश्च शोणितातिस्त्रवाच्च यः॥
अभिघातनिमित्तश्च न सिद्धयत्यपतानकः।

Ma.Ni. – 22/38

The following parameters met with the patient of *Apatanaka* is considered as *Asadhya Vikara* (incurable).

- The development of Apatanaka in the lady during the pregnancy
- When Apatanaka disease is manifested due to excessive bleeding (excessive Rakta Pravritti) from the body[9]
- If the individual suffering from Apatanaka, manifested in him due to any physical trauma

GK

7 This spectrum of the diseases is attributable to the seizure episodes which manifest due to the presence of organic lesion such as intracranial space occupying lesion (brain tumour etc). Whereas seizure episodes without the presence of organic (structural) lesion is termed as Apasmara. Thus, the difference between Akshepaka and Apasmara is the presence and absence of organic (structural lesion) respectively.

8 The clinical features mentioned in the Dandaka spectrum of diseases is attributable to the features of status epilepticus. Status epilepticus is defined as a seizure that lasts longer than 5 minutes or having more than 1 seizure within a 5 minutes period, without returning to a normal level of consciousness between episodes. This disease is a medical emergency, if untreated, the person may have permanent brain damage or death.

- There are several risk factors for development of status epilepticus. These include poorly controlled epilepsy, low blood sugar, cerebrovascular accident, renal failure, hepatic failure, encephalitis, HIV, alcohol or drug abuse, head injuries

9 This can be attributable to the clinical features of eclampsia. This disease is a rare but serious complication of pre eclampsia. Pre-eclampsia is a disorder of pregnancy in which the lady with this disease develops seizures (convulsions) during pregnancy. Seizures are episodes of shaking, confusion and disorientation caused by abnormal brain activity. This usually occurs after 20th week of pregnancy. This is rare and affects < 3% of people with pre-eclampsia.

Apatantraka[10]

क्रुद्धः खैः कोपनैर्वायुः स्थानादूर्ध्वं प्रपद्यते।
पीडयन् हृदयं गत्वा शिरः शङ्खौ च पीडयन्॥12॥
धनुर्वन्नमयेद्गात्राण्याक्षिपेन्मोहयेत्तथा।
(नेमयेचाक्षिपेचाङ्गान्युच्छासं निरुणद्धि च॥)
कृच्छ्रेण चाप्युच्छ्वसिति स्तब्धाक्षोऽथ निमीलकैः॥
कपोत इव कूजेच्च निःसंक्षः सोऽपतन्त्रकः।
दृष्टिं संस्तम्भ्य संज्ञां च हत्वा कण्ठेन कूजति॥14॥ Ca.Ci. – 9/12

- Vitiated *Vata Dosha* will move upwards from its normal location, thereby leads to the affliction of *Hridaya*, *Shira* and *Shankha* region, due to this, the person may have bending of the body in a similar fashion as the bending of the bow, and it is associated with frequent involuntary movements.
- This is associated with breathlessness during expiration, stiffness of the *Indriya*, closing of the eyes, sound produced by the patient is simulating the sound produced by pegion, altered sensation, fixed gaze and sound is produced from the throat region.

क्रुद्धः स्वैः कोपनैर्वायुः स्थानादूर्ध्वं प्रपद्यते॥28॥
पीडयन् हृदयं गत्वा शिरःशङ्खौ च पीडयन्।
धनुर्वन्नमयेन्नात्राण्याक्षिपेन्मोहयेत्तदा॥29॥
स कृच्छ्रादुच्यसेच्चापि स्तब्धाक्षोऽथ निमीलकः।
कपोत इव कूजेच्च निःसंशः सोऽपतन्त्रकः॥30॥ Ma.Ni. – 22/28-30

- Because of the vitiation of *Vata Dosha* in the body due to various *Nidana* (aetiolgical factors mentioned in *Ayurveda* textbooks), abnormally increased *Vata Dosha* move upwards and cases pain in the chest region, head and temporal region.
- This leads to the abnormal bending of the body like a bow, associated with involuntary movements of the body and altered state of consciousness. The patient will have difficulty in breathing, eyes wide open without movement of lids or eyes completely closed. In this condition, the patient voice may have the production of sounds which simulates the sound of pigeon.

Dandapatanaka

कफान्वितो भृशं वायुस्तास्वेव यदि तिष्ठति॥
दण्डवत्स्तम्भयेद्देहं स तु दण्डापतानकः। Ma.Ni.. – 22/32

- When vitiated *Vata* leads to the stiffness of the hands, feet, head, back and hip, the becomes stiff and the stiffness resembles stick (*Danda*), hence the name *Dandapatanaka*[11] as per *Sushruta* and *Madhavakara*. This is also being termed as *Dandaka* as per *Charaka*.

Antarayama[12] or Abhyantara Ayama

Ma.Ni. – 22/33

धनुस्तुल्यं नमेद्यस्तु स धनुस्तम्भसंज्ञकः॥
अङ्गुलीगुल्फजठरहृद्वक्षोगलसंश्रितः।
स्नायुप्रतानमनिलो यदाऽऽक्षिपति वेगवान्॥34॥
विष्टब्धाक्षः स्तब्धहनुर्भग्नपार्श्वः कफं वमन्।
अभ्यन्तरं धनुरिव यदा नमति मानवम्॥35॥
तदाऽस्याभ्यन्तरायामं कुरुते मारुतो बली। Ma.Ni. – 22/34-36

- When vitiated *Vata* obstructs Manya region situated in the neck region the *Dhamani* get pulled inside, making the body to be bent like *Dhanush* (bow) from the anterior side.
- This is associated with stiffness of the eyes, yawning, biting of the teeth, vomiting of mucus, pain in the flanks, stiffness of the neck and stiffness of the head.

Bahirayama[13]

GK

10 Apatanaka and Apatantraka are the spectrum of diseases which manifest as a complication of Hridroga.
हृद्रोगमप तन्त्रकमाह ! Ca.Si. – 9/12-15, Chakra
- This spectrum of diseases can be attributable to development of altered state of consciousness or involuntary movements manifested due to the primary source of pathogen present in the cardiac region.
- In this occasion, due to either increased viscosity of the blood or due to thrombosis formed in the heart getting embolized into the cerebral vessel or due to the atherosclerotic changes in the ascending aorta which get embolized into the cerebral vessel, the person may develop the features of altered consciousness. This may be associated with involuntary movements when the blood supply to the basal ganglia or midbrain or cerebellum is getting hampered or completely stopped.

11 Refer to the context of Dandaka mentioned in this chapter.

12 The clinical features of Antarayama spectrum of diseases are attributable to decorticate lesion of the central nervous system.
- The term decorticate lesion refers to an abnormal posturing in which a person is stiff with bent arms, clenched fists, and legs held out of straight. The arms are bent towards the body and the wrists and fingers are bent and held on the chest.
- This is indicative of sign of severe damage in the brain.
- This is indicative of damage to the nerve pathway in the midbrain, which is between the brain and spinal cord.
- Although, decorticate lesion or posture is serious, it is usually not as serious as a type of abnormal posture called decerebrate posture.

13 The clinical features of Bahirayama spectrum of diseases are attributable to decerebrate lesion of the central nervous system.
- Decerebrate posture is an abnormal body posture that involves the arms and legs being held straight out, the toes being pointed downward, and the head and neck being arched backward. The muscles are tightened and held rigidly. This type of posturing usually means there has been severe damage to the brain.
- A severe brain injury is usual cause of this type of posture.
- This posture can occur unilaterally or bilaterally. It can occur just in the arms also. This may alternate with decerebrate posture.
- The diseased individual may have decorticate lesion on one side of the body and decerebrate lesion on the other side.
- Bleeding in the brain from any cause; brain stem tumours; cerebrovascular accidents; traumatic brain injury; brain dysfunction due to illicit drugs, poisoning or infection; intracranial space occupying lesion (ICSOL); brain injury due to hypoxia can cause decerebrate

बाह्यस्नायुप्रतानस्थो वाह्यायामं करोति च॥36॥
तमसाध्यं बुधाः प्राहुर्वक्षःकट्युरुभञ्जनम्। Ma.Ni. – 22/36,37

- When vitiated *Vata* leads to the bending of the body, the nature of bending will be like bending of *Dhanush* (bow) from the posterior side. The *Snayu* of the chest, pelvic region undergo stiffness of the body along with bending posteriorly. This is considered as *Asadhya* (incurable) in nature.
- This is associated with gnashing of the teeth, yawning, salivation and loss of speech, when it comes with force, it kills the patient or produces restlessness.

Ardita

खस्थः स्यादर्दितादीनां मुहुंर्बेगे गतेऽगते॥ Ca.Ci. – 28/52

Ardita is the spectrum of diseases, which is explained in different contexts. As per the descriptions available in *Ayurveda* literatures, appearance of clinical features, duration of the illness determines the diagnosis and prognosis of *Ardita*. Literally, the term *Ardita* refers to the illnesses which afflict the individuals in episodes. In other words to say, when we want to do the diagnosis of Ardita, he will have symptoms only during the episodes of the illness and symptoms will be remitted or subsided in between the episodes of the illness. As per *Charaka*, *Ardita, Antarayama, Bahirayama, Hanustambha, Akshepaka* and *Dandaka* are included under the large umbrella of clinical spectrum *Ardita*. As per *Charaka*, *Ardita* can afflict the face or face and one side of the body. *Chakrapani Datta*, commentator of *Charaka Samhita* mentioned the difference between *Ardita* and *Pakshaghata* as these two illnesses overlap in clinical presentation. *Ardita* is observed as a episodic illness and *Pakshaghata* is a persistent illness and never occurs as episodic illness. As per *Susrhruta, Ardita* can afflict only face of the individual. Affliction of one part of the body can be included in *Pakshaghata* spectrum. But, for the practical understanding, *Ardita* is the illness which afflicts the face of the individual barring the other body parts.

Nidana of Ardita

गर्भिणीसूतिकाबालबृद्धक्षीणेष्वसृक्षये॥
उच्चैर्व्याहरतोत्यर्थ खादतः कठिनानि च॥68॥
हसतो जृम्भतो भाराद्विषमाच्छ्रुसनादपि॥ Su.Ni. – 1/68,69

- When an individual is having excess *Rakta Kshaya* (blood loss from the body) such as pregnancy[14], puerperal period[15], children[16], elderly people[17], people who are having excess depletion of body tissues[18]
- When the person is shouting persistently at the top of his voice
- When the person is eating very hard food articles[19]
- Excess laughing[20]
- Excess yawning[21]
- Excess carrying of heavy weight objects on head[22]
- Two different meanings are there for this term, lying on uneven ground or surfaces or in case of abnormal lesion.

GK

14 When a pregnant lady is presented with seizures, hypertension and proteinuria, she is likely to have cerebrovascular accident, if there is blockage of the blood supply to the brain or surroundings of the brain secondary to hypertension. This may occur as a result of either infarction due to thromboembolism or due to haemorrhage due to hypertension.

15
- Risk factors for stroke during pregnancy and the puerperal period include gestational hypertension, HELLP syndrome (characterized by hemolysis, elevated liver enzymes, and thrombocytopenia), frequent vomiting, and changes in hemolysis and the pre-thrombotic state in late pregnancy and the postpartum period
- age ≥40 years, black and Asian races, hemorrhagic stroke (compared with ischemic stroke), anemia, heart failure, cardiomyopathy, atrial fibrillation, hypertension, preeclampsia/eclampsia, gestational diabetes, and cesarean delivery

16 AV malformation in brain in children is one of the major cause of the manifestation of neurological manifestation.

17 History of coronary arterial disease (CAD), hypertension, diabetes mellitus, dyslipidaemia, intracerebral aneurysms are the possible risk factors and possible causes of neurological manifestation in elderly people.

18 When an individual is consuming less amount of nutritious food, he may likely to develop nutritional deficiency which in turn leads to the conversion of stored fat into circulating fat. When there is conversion of stored fat into circulating fat, there is a risk of atherosclerosis in the aorta or common carotid vessels or in the intracerebral vessels. Thus, the person may develop neurological manifestation. This can be considered as one of the two major mechanisms of the Vatavyadhi – Dhatu Kshaya, the other one is being Avarana.
- Avarana mechanism of Vatavyadhi is considered to be occurring as Santarpana type of pathophysiology. Dhatu Kshaya is considered to be acting as Apatarpana type of pathophysiology as per Ayurveda fundamentals as far as the Samprapti of Vatavyadhi is concerned.

19 It is being said that upper respiratory ailments as one of the risk factors for the development of neurological manifestation, Bell's palsy in particular. When an individual is consuming hard substances, it may lead to improper mastication of food, which in turn may lead to swallowing difficulties, which in turn may lead to aspiration of food either to the nose or to the larynx, trachea etc. This may lead to the manifestation of upper respiratory ailments, if the food enters the nose and larynx.

20 pathological laugher is considered as one of the important prodromal symptoms of cerebrovascular accident. It results from the impairment in the connectivity between frontal cortex, temporal cortex, hypothalamus, amygdala, and cerebellum, which are involved in various steps of modulation of laughter phenomenon.

21 Pathological yawning is one of the clinical sign in disorders affecting the brainstem. This can be seen in the patients of acute MCA (middle cerebral artery stroke) and in supratentorial stroke. Migraine, epilepsy, disorders of basal ganglia, multiple sclerosis and multiple sclerosis are responsible for the manifestation of pathological yawning.

22 It is being said that normal cerebral blood flow is around 50 ml/ 100gm of brain/ minute. Carrying heavy load on one's head may likely to increase the blood flow to several fold. Thus, the person may have development of intracerebral haemorrhage if this habit is frequently performed.

breathing patterns[23]

Lakshana of Ardita

शिरोनासौष्ठचिबुकललाटेक्षणसन्धिगः॥69॥
अयित्वाऽनिलो वक्रमर्दितं जनयत्यतः॥
वक्रीभवति वक्रार्धे ग्रीवा चाप्यपवर्तते॥70॥
शिरश्चलति वाक्सङ्गो नेत्रादीनां च वैकृतम्॥
ग्रीवाचिबुकदन्तानां तस्मिन् पार्श्वे तु वेदना॥71॥
यस्याग्रजो रोमहर्षो वेपथुर्नेत्रमाविलम्॥
वायुरूर्ध्व त्वचि स्वापस्तोदो मन्याहनुग्रहः॥
तमर्दितमिति प्राहुर्व्याधिं व्याधिविशारदाः॥72॥ Su.Ni. – 1/69-72

अतिवृद्धः शरीरार्धमेकं वायुः प्रपद्यते।
यदा तदोपशोष्यासुग्बाहुं पादं च जानु च॥38॥
तस्मिन् सङ्कोचयत्यधै मुखं जिह्मं करोति च।
वक्रीकरोति नासा भूललाटाक्षिहनूस्तथा॥39॥
ततो वक्रं व्रजत्यास्ये भोजनं वक्रनासिकेंम्।
स्तब्धं नेत्रं कथयतः क्षवयुश्च निगृहाते॥40॥
दीना जिल्ला समुत्क्षिप्ता कला सज्जति चास्य वाकू।
दन्ताश्चलन्ति वाध्येते अवणौ भिद्यते स्वरः॥41॥
पादैहस्ताक्षिजलेरुशङ्खश्रवणगण्डरुकू।
अर्धे तस्मिन्मुखार्थे वा केवले स्यात्तदादितम्॥42॥

Ca.Ci. – 28/38-42

- Affliction of head, nose, lips, chin, forehead, periorbital region takes place leading to Ardita
- Deviation of mouth to one side
- Stiffness of the neck[24,25]
- The head keeps on shaking[26]
- Slurred speech[27]
- Asymmetry of the eyes[28]
- Discomfort in the nape of neck, chin and teeth, horripulation, involuntary movements[29]
- Numbness of the skin
- Pricking type of pain[30]
- Stiffness of nape of neck
- Stiffness of the temporo mandibular joint[31]
- Ingested food comes through the nose[32]
- Weak teeth[33]

Figure 165: Lakshana of Ardita

GK

23 It is being said that sleep disorders including sleep disordered breathing, insomnia, hypersomnia, parasomnias and sleep related movement disorders are linked with the manifestation of cardiovascular conditions, which in turn increases the risk of development of cerebrovascular events.

24

25 In Bell's palsy (LMN facial nerve palsy), there will be deviation of mouth towards the sound side. This means on the sound side of the face, increased muscular tension is noticed, if this is continuously observed in the patient of Bell's palsy, there is a possibility of development of contractures of the muscles of the neck. This is leading to stiffness of the neck.

26 In western medicine, this is being termed as titubation. It means a staggering gait or swaying or shaking of the trunk or head that is observed especially in individuals affected with a disease of cerebellum.

- It is being said that facial nucleus i.e nucleus of VII cranial nerve which is located in pons gets its blood supply from anterior inferior cerebellar artery (AICA) which is a branch of basilar artery. T
- Thus, any blood supply is hampered from this artery, facial nerve nucleus may not get adequate blood supply, thus, this may be a clinical presentation in the patient of VII cranial nerve palsy.

27 As articulation of speech is affected, the person likely to have slurred speech. Words starting from p, b, m require movement of lips; words starting from t, d etc require palatal movement. In the person suffering from VII cranial nerve palsy, this may be affected, thus he may develop slurred speech.

28 This is specifically found in LMN variety of facial palsy. There will be inability of the patient to completely close the eyes is being observed. Thus, asymmetry in the appearance of the eyes is likely.

29 Hemifacial spasm is a disorder characterized by paroxysmal, involuntary twitching of facial muscles of one side of the face innervated by the ipsilateral facial nerve (VII cranial nerve). It is considered as one of the subtype of peripheral (neuromuscular) movement disorder.

30 Facial nerve is considered to be mixed nerve which has both sensory and motor functions. Sensory function of the facial nerve supplies nerve supply to external auditory canal, tympanic membrane, pinna of the ear, and anterior 2/3rd of the tongue. When the person is suffering from facial nerve palsy, these areas may have either numbness or sensory impairment in these regions.

31 Refer to the description mentioned in the context of stiffness of the neck in this context of this chapter.

32 This is attributable to dysfunction of soft palate. In this occasion, the person will not have complete closure of nasal cavity occurs during swallowing, which may lead to aspiration of food into the nasal cavity, resulting in the coming of the food through the nose.

33 This is attributable to loss of muscle tone of oral muscles of the affected side, where the mastication of the food will be decreased.

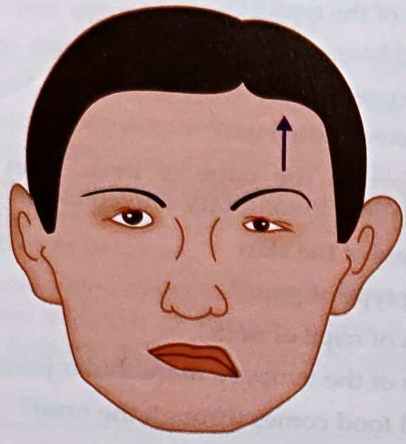

Figure 166: Hemifacial spasm

- Hearing impairment[34]
- Hoarseness of voice[35]
- Pain in the foot, palm, eyes, thigh, temporal region, ears and cheeks[36]

Sadhyasadhyata of Ardita

क्षीणस्यानिमिषाक्षस्य प्रसकाव्यक्तभाषिणः॥
न सिध्यत्यर्दितं वार्ड त्रिवर्ष वेपनस्य च॥73॥ Su.Ni. – 1/73

- If the person suffering from *Ardita* is emaciated[37], *Asadhya* (incurable).
- If the eye of the person is fixed on gaze[38], then *Asadhya* (incurable).
- If one is suffering from irrelevant or incomprehensible speech[39] continuously, *Asadhya* (incurable).
- If the duration of illness is long[40], *Asadhya* (incurable)
- If the duration of the illness is of *Tri Varsha* (3 years duration), *Asadhya* (incurable) which indicates the chronicity of the illness.
- If the person of *Ardita* has secretions from *Nasa* (nostril), *Akshi* (eyes) and *Mukha* (oral cavity), then *Asadhya* (incurable).
- If the person is having involuntary movements[41], then *Asadhya* (incurable).

Pakshaghata

गृहीत्याऽर्थ तनोबांयुः सिराः नायुर्विशोष्य च॥39॥
पक्षमन्यतरं इन्ति सन्धिवन्धान्विमोक्षयन्।
कृत्नोऽर्घकायस्तस्य स्यादकर्मण्यो विचेतनः॥40॥
एकाङ्गरोगं तं केचिदन्ये पक्षवधं विदुः।
सर्वाङ्गरोगस्तद्धश्च सर्वकायाधितेऽनिले॥41॥ Ma.Ni. – 44/39-41
हत्यैकं मारुतः पक्षं दक्षिणं वाममेव वा॥53॥
कुर्याचेष्टानिवृत्ति हि रुजं वाफ्स्तम्भमेवं च।
गृहीत्वाऽधे शरीरस्य सिराः स्त्रावृर्विशोष्य च॥54॥
पादं संकोचयत्येकं हस्ते वा तोदशूलछत्।
एकाङ्गरोगं तं विद्यात् सर्वाङ्गं सर्वदेहजम्॥55॥ Ca.Ci. – 28/53-55

- Vitiated *Vata Dosha* occupies and affects half of the

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34 Vestibulo cochlear nerve (VII cranial nerve) is responsible for normal hearing. This may be usually affected along with facial nerve (VII cranial nerve) affliction. In this occasion, the person may have hearing impairment, tinnitus and dizziness. This is presented with herpetic vesicles around pinna or oral mucosa, otalgia and facial nerve palsy.
- We can have another hypothesis to be put forth in case of hearing impairment. When the person is presented with regurgitation of food from oral cavity into nasal cavity due to improper closure of soft palate. This may lead to the occlusion of eustachian tube which connects between middle ear and nasal cavity. Due to this, the air in the middle ear is absorbed by the surrounding tissues, thereby the person may have conductive variety of deafness.

35 This may be attributable to the aspiration of the food into the larynx due to improper functioning of soft palate, tongue etc, which may be identified by hoarseness of voice.

36 This type of clinical presentation is observed in the patiehts where whole half of the body is afflicted including face of the individual.

37 Vatavyadhi is manifested in individuals either due to Avaravana (Santarpana effect) and Dhatu Kshaya (Apatarpana effect). Santarpana effect can be seen in the patients who are having the habit of over energy intake; whereas Apatarpana effect can be seen in the patients who are taking less amount of nutritious diet, this may lead to emaciation of the body which ocurrs as a result of conversion of stored fat into circulating fat.
- Nutrition requirement of brain: Normal adult requires 200 gms of glucose per day; two-thirds of which (130 gms) is specifically needed by the brain to cover its glucose needs. The brain competes with rest of the body for glucose when levels dip very low – such as during starvation.
- Thus Apatarpana Nidana causes nutrient deprivation to the brain, thereby the person may develop abnormal neurological manifestation.

38 There is one condition called gaze palsy. The term gaze palsy refers to sudden restricted movement of the eyeball. This is specifically observed in midbrain lesions. When there is involvement of midbrain, it is considered to be difficult to subsidence of the symptoms completely.

39 As far as disturbed speech is concerned, dysphasia and dysarthria are present in the patients of Ardita. Dysphasia is considered to be occurring as a result of brain dysfunction and dysarthria is resulted when the patient is having disturbances in articulation of muscles of speech.
- Further, the person may have fluent and non fluent speech. Fluent speech involves normal or near normal speech, but incomprehensible speech, usually seen in parietal lobe lesions. Non fluent speech involves less amount of irrelevant and incomprehensible speech, which may be seen in frontal lobe lesions.

40 It is being said that shorter illness requires shorter line of treatment for recovery; long duration illness requires long duration of treatment.

body. This abnormal *Vata* is carried through *Sira* and dries up the *Snayu*.

- Clinical manifestation is loss of strength on half[42] of the side either right or left side.
- Pricking type of pain on affected side of the body[43]
- Altered state of consciousness[44]
- Associated with pain in the affected part[45]
- Speech disturbances[46]
- This is also being termed as *Ekangaroga* and sometimes *Pakshavadha*.
- When similar mechanism involves whole body, that is termed as *Sarvangaroga*.

अधोगमाः सतिर्यग्गा धमनीरुर्ध्वदेहगाः॥
यदा प्रकुपितोऽत्यर्थं मातरिश्वा प्रपद्यते॥60॥
तदान्यतरपक्षस्य सन्धिबन्धान् विमोक्षयन्॥
हन्ति पक्षं तमाहुर्हि पक्षाघातं भिषग्वराः॥61॥
यस्य कृत्वं शरीरार्धमकर्मण्यमचेतनम्॥
ततः पतत्यसुन् वापि जहात्यनिलपीडितः॥62॥ Su.Ni. – 1/60-62

- When abnormally vitiated *Vata Dosha* gets localised in *Dhamani* going downward, obliquely and upwards, it paralyses single side of the body.
- *Dushta* or *Prakupita Vata* dissociates the joints between *Sira, Snayu* and *Kandara*.
- Abrupt loss of movements in one half of the body

Sadhyasadhyata of Pakshaghata[47]

शुद्धवातहतं पक्षं कृच्छ्रसाध्यतमं विदुः॥
साध्यमन्येन संसृष्टमसाध्यं क्षयहेतुकम्॥63॥ Su.Ni. – 1/63

- Single *Vata* dominant *Pakshaghata*[48] is considered as *Krichrasadhya* (difficult to cure)
- If association of other *Dosha*[49] along with vitiated *Vata* is considered to be *Sadhya* (easily curable)
- If the *Pakshaghata* is manifested due to *Dhatu Kshaya*[50], then such type of *Pakshaghata* is *Asadhya* (incurable) in

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42 This type of clinical presentation is seen in the patients of hemiparesis or hemiplegia. The lesions present in midcervical spinal cord and above are responsible for this type of clinical presentation. The presence of other associated features helps in the localization of the lesion.
- Language disorders, cortical sensory disturbances, cognitive abnormalities, disorders of visual-spatial integration, apraxia or seizures reflect a cortical lesion.
- Homonymous visual field defects reflect either a cortical or a subcortical hemispheric lesion.
- A pure motor hemiparesis of the face, arm or leg is often due to a small, discrete lesion in the posterior limb of internal capsule, cerebral peduncle or upper part of pons.
- Crossed paralysis consisting of ipsilateral cranial nerve signs and contralateral hemiparesis implies the diagnosis of brainstem lesions.
- The absence of cranial nerve signs or facial weakness suggests that a hemiparesis is due to a lesion in the high cervical spinal cord, especially if associated with ipsilateral loss of proprioception and contralateral loss of pain and temperature sense (Brown Sequard syndrome).

43 This is considered to be neuropathic pain. This is caused by damage to the brain's pain processing pathways, rather than because of injury. It is also called central post stroke pain (CPSP). This type of pain occurs more often when sensation is reduced after a stroke. The brain is used to receiving normal sensory inputs, and when it doesn't, the brain itself produces painful sensations.
- CPSP may feel like burning, stabbing, pricking or numbness on the skin. It mostly occurs on stroke affected side of the body. Often, this pain is made worse if affected part is touched or moved or affected part is placed in water. CPSP may start days, months or years after the manifestation of stroke.

44 This may be observe in the patient who has attained cerebral oedema either due to ischemia or haemorrhage. This may also be in the state of electrolyte imbalance.

45 This can be attributable to either neuropathic pain or developed due to manifestation of contractures.

46 This can be attributable to either dysphasia (where the lesion seen in brain) or dysarthria (where the lesion is seen in muscles of articulations of speech).

47 Pathological events of cerebrovascular accidents (CVA) or strokes are initially the person will not have development of UMN signs and at that time, the person will have flaccidity which mimics the features of spinal shock. In due course of time, the person will have UMN signs.

48 The term Vata dominant Pakshaghata refers to the duration of the illness is prolonged. During this period, adequate nutrition to the main affected area will be less, unless blood supply to the afflicted part of the brain takes place within a period of 4 minute. This is associated with the presence of increased UMN signs such as hypertonia, hyperreflexia, development of contractures etc.

49 This can be observed in the initial phase of cerebrovascular accident wherein the person may have either sensory impairment such as burning sensation or heaviness of the affected body part. This may suggest the involvement of Pitta or Kapha along with vitiated Vata Dosha in the body. This implies the duration of the illness is shorter when compared to the Pakshaghata of vitiated Vata Dosha only. Hence, this type of association of Dosha, i.e. Samsargaja Dosha state is easily curable.

50 Ca.Ci. – 28/59
Chakra, Ca.Ci. -28/59
As per Charaka, Vatavyadhi is manifested due to two different mechanisms – Avarana and Dhatukshaya. In most of the occasions, Avarana occurs because of Santarpana type of causes and Dhatukshaya occurs because of Apatarpana type of causes. Chakrapani opines Dhatukshaya as Sara Kshaya i.e. loss of essential nutrients from the body.
- As per current understanding, whenever there is deficiency of nutrients either due to poor absorption or due to poor circulation, there will be deficiency of essential macronutrients (carbohydrates, proteins, fats) and micronutrients (sodium, potassium and calcium ions) and water loss.
- Carbohydrate, one of the essential macronutrient, electrolytes (sodium, potassium, calcium) and water is necessary for almost all activities of skeletal, smooth and cardiac muscles and all kinds of neuronal function. When there is deficiency of these substances, naturally

nature

Kampavata

सर्वाङ्गकम्पः शिरसो वायुर्वेपथुसंज्ञकः। Ma.Ni. – 22/74

Kampavata is the spectrum of diseases where the individual may develop involuntary movements[51] in whole body or involuntary movement of the head. It is also being termed as *Vepathu*. This is manifested in the individual due to the vitiation of *Vata Dosha* in the body. As per *Madhukosha* commentary, involuntary movement may be observed in the upper and lower limbs altogether or separately. It can be observed as unilateral or bilateral presentation. When we speak about the *Amshamsha Kalpana* of *Dosha*, it is the increase of *Chala* (mobile) quality of *Vata Dosha* is seen during the progression of the disease.

Gridhrasi

स्फिक्पूर्वा कटिपृष्ठोरुजानुजङ्घापदं क्रमात्।
गृध्रसी स्तम्भरुक्तोदैर्युक्वाति स्पन्दते मुहुः॥56॥
बाताद्वातकफात्तन्द्रागौरवारोचकान्विता।

Ca.Ci. – 28/56,57; Ma.Ni. – 22/54

डपातंज्ञायां भवेत्तोदो देहस्यापि प्रवकता (?)।
जानुकट्पुरुसंधीनां स्फुरणं स्तब्धता भृशम्॥55॥
वातश्लेष्मोद्भवायां तु निमित्तं वह्निमार्दवम्।
तन्द्रामुखप्रसेकश्च भक्तद्वेषस्तथैव च॥56॥ Ma.Ni. – 22/55,56
पाणीप्रत्यङ्गुलीनां तु कण्डरा याऽनिलार्दिता॥
सनोः क्षेपं निगृह्णीयाद् गृधेसीति हि सा स्मृता॥ Su.Ni. – 1/74

- *Vitiated Vata Dosha* either singly or associated with *Kapha Dosha* will be responsible for the manifestation of the symptoms of Gridhrasi[52].
- This disease is characterised by stiffness of the lower limb and severe pain commences from lower back radiating downwards through buttocks, thighs, knees, calves and feet.
- The discomfort is experienced intermittently
- When there is association of *Kapha Dosha*, the disease is characterised by discomfort of *Vataja Gridhrasi* along with drowsiness, heaviness of the lower part of the body and tastelessness.
- There will be bending of the body due to the radiation of the pain to the lower limb.
- As per *Madhava Nidana*, there will loss of appetite in case of *Vatakaphaja Gridhrasi*.

The person is unable to lift the lower limb without bending the knees.

Vishvachi

तलप्रत्यङ्गुलीनां तु कण्डरा बाहुष्पृष्ठतः॥
बालोः कर्मक्षयकरी विश्यांचीति हि सा स्मृता॥75॥ Su.Ni. – 1/75
तैलं प्रत्यङ्गुलीनां या फण्डरा बाहुपृष्ठतः।
बीडुचेष्टापहरणी विश्वाची नाम सा स्मृता॥44॥ A.Hr.Ni. – 15/44

- This disease[53] is mainly caused by vitiation of *Vata Dosha* in the upper limb.
- In this, *Kandara* of palm, fingers, arm and shoulder of upper limb is afflicted
- As per *Dalhana*, the clinical features of this disease is similar to the pain of Gridhrasi of lower limb.
- The person will be having pain observed in the neck radiating to whole upper limb till fingers.

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the person may have poor neuronal function and that too, in a person who is suffering from Pakshaghata, adequate supply of these elements is necessary, which is hampered in the patients of Pakshaghata, hence such patients of Pakshaghata is incurable in nature.

51 Whenever involuntary movements observed in the human body, exact distribution and type of the involuntary movement must be recorded. The following are different names of involuntary movements named on the specific patterns observed in them.
- Chorea: repetitive, jerky movements which are irregular, spontaneous, purposeless and may occur anywhere in the body, precipitated by emotion and voluntary effort and disappears during sleep.
- Athetosis: peculiar, slow snake charming movements seen chiefly affecting wrists and fingers. Flexion and extension are the movements observed in this type.
- Hemiballismus: a swinging movement of violent nature of arm and leg of great amplitude at shoulder and hip, but affected limbs are flaccid.
- Tics and habit spasms: these are difficult to distinguish from chorea. Here, the movements are purposive. Some are simple such as shrugging shoulders, blinking eyes and some are complex such as frequent uncontrollable repetition of same words or sentence. These movements may go on for longer duration than chorea i.e. months without much variation in intensity.
- Tremors: these may occur due to metabolic derangement, alcoholism. Tremors of neurological origin are of 2 kinds: static and action
  - o Static tremor: relatively independent of movement e.g. 3-5/ second course tremor in case of parkinsonism; first affects index and thumb of one hand, followed by flexion-extension of wrist followed by pronation – supination of forearm.
  - o Action tremor: this is also called intention tremor, seen in ipsilateral cerebellar lesion.
- Myoclonus: a rare type consists of sudden shock like contraction producing recurring muscle jerks.

52 This is the spectrum of the diseases where the person will be having the presence of pain in the lower limb. This type of clinical feature is seen in the patients of lumbar spondylosis where degenerative lesions are compressing the spinal nerves or in the patients suffering from intervertebral disc prolapse in the lumbar region or in the conditions where increased intraspinal pressure such as intraspinal tumours are seen. The pain gets worsened on coughing, sneezing or straining for defaecation or any maneuvers which increase the intraspinal pressure.

53 This is the spectrum of the diseases is responsible for the presence of pain in the upper limb. This type of clinical feature is seen in the patients of cervical spondylosis or the conditions where increased intraspinal pressure is seen such as intervertebral disc prolapse in cervical region, intraspinal tumour afflicting the cervical region etc. The pain gets worsened on coughing, sneezing or straining for defaecation or any maneuvers which increase the intraspinal pressure.

- This is associated with reduced functioning of the upper limb also.
- The combination of *Vishvachi* and *Gridhrasi* is called as *Khalli*. *Khalli* is characterised by pain, crippling and contracture of both upper and lower limbs.

Panguta

बायुः कठ्यां खितः सक्ष्नः कण्डरामाक्षिपेयदा।
खजस्तदा भवेजन्तुः पेङ्गः सनोद्ब्योर्षधात्॥77॥ Su.Ni. – 1/77

- Vitiated *Vata Dosha* takes shelter in the Kati region and afflicts the *Kandara* of *Sakthi* (thigh) region and this is responsible for the loss of function of lower limb either one lower limb or both lower limbs[54].
- It can occur in one lower limb or both lower limbs.
- If the loss of strength in one lower limb is observed, it is being termed as *Khanja*[55].
- If the loss of strength of both lower limbs is observed, it is being termed as *Pangu*[56].

Difference between Khanja and Gridhrasi

Khanja Gridhrasi

- Complete loss of functioning of one limb occurs pain, restricted movement of the affected part of the body, no complete loss of functioning
- *Vata* dominant disease *Vata* or *Vata-Kapha* dominant disease
- No radiation of pain in the lower limb radiation of pain from buttock to lower limb

To summarise

- To summarise the concept of *Snayugata Vata*, clinical features such as *Stambha* (stiffness), *Sankocha* (contractures), *Khalli* (muscular cramps), *Granthi* (nodular swellings), *Sphurana* (irregular pulsations) are being observed.
- *Stambha* can be observed if the illness is of short duration and if there is involvement of *Asthi* and *Sandhi* etc.
- *Sankocha* is observed in case of long standing illnesses associated with involvement of *Mamsa Dhatu* as well. *Khalli* is observed when there is extreme loss of electrolytes from the body.
- *Granthi* can be observed in case of involvement of myelin sheath. *Sphurana* can be observed in case of fasciculation which is one of the LMN signs observed in case of neurological deficit.

54 Normal motor function involves muscle activity that is mediated by the activity of cerebral cortex, basal ganglia, cerebellum and spinal cord. Motor dysfunction in any of the level may lead to the presenting complaint of either reduced strength or complete loss of the strength on the affected part. In this occasion, it is the affliction of lower limbs is seen.

55 This spectrum of the disease can be considered as monoparesis or monoplegia of one lower limb. This usually occurs due to the lower motor neuron disease which supplies lower limbs with or without associated sensory involvement. UMN weakness occasionally presents as monoparesis of distal or nonantigravity muscles. It can occur either suddenly or gradually.

- Acute monoparesis: If the weakness is predominantly in distal and nonantigravity muscles and not associated with sensory involvement or pain, focal cortical ischaemia is likely to be diagnosed.
  - o Sensory loss and pain usually accompany acute lower motor neuron weakness; the weakness is commonly localized to a single nerve root or peripheral nerve within the limb but occasionally reflects plexus involvement.
  - o If LMN weakness is suspected, or pattern of weakness is uncertain, the clinical approach begins with EMG and nerve ccondutction study.
- Subacute or chronic monoparesis: weakness and atrophy that develop over weeks or months are usually of LMN origin.
  - o If they are associated with sensory symptoms, a peripheral cause (nerve, root or plexus is predicted. In the absence of such symptoms, anterior horn cell disease is predicted. Here, EMG is performed to confirm the diagnosis.
  - o If weakness is of UMN type, d discrete cortical (precentral gyrus) or cord lesion may be predicted, appropriate imaging is done to confirm the diagnosis

56 This spectrum of the disease can be termed as paraparesis or paraplegia. Intraspinal lesion at or below upper thoracic spinal level, as most observed pathological event. But, paraparesis may also occur from lesions at other locations that disturb upper motory neurons (parasagittal intracranial lesions); and lower motor neurons (anterior horn disorders, cauda equina syndromes due to involvement of nerve roots derived from the lower spinal cord. It can occur either suddenly or gradually.

- Acute paraparesis is little bit difficult to diagnose initially due to spinal cord disease if the lower limbs are having flaccidity and areflexia.
  - o Sensory loss in the legs with an upper level on the trunk; a dissociated sensory loss suggestive of a central cord syndrome; or exaggerated reflexes in the legs with normal reflexes in the arms.
  - o Compressive lesions (epidural tumour, abscess or haematoma), but also a prolapsed intervertebral disc and vertebral involvement by malignancy or infection, transverse myelitis are some of the acute causes of paraparesis.
  - o Diseases of anterior cerebral artery ischaemia produce acute paraparesis with disturbed shoulder shrugging, superior sagittal sinus or cortical venous thrombosis and acute hydrocephalus may also lead to acute paraparesis. If upper motor neuron signs are associated with drowsiness, confusion, seizures or other hemispheric signs, MRI brain should be done to confirm the diagnosis.
  - o Paraparesis may result from cauda equina syndrome (e.g. following trauma to the low back, a midline disc herniation or intraspinal tumour).
- Chronic paraparesis with spasticity is caused by UMN disease. When there is associated with lower limb sensory loss with sphincter involvement, a chronic spinal cord disorder is likely. If MRI of spinal cord is normal, then MRI brain has to be done.
  - o When hemispheric signs are present, then a parasagittal meningioma or chronic hydrocephalus is suspected.

Key summary of the chapter

- Akshepaka is the condition where involuntary movements are seen. This is categorized under Vatavyadhi. It can be attributable to epilepsy due to organic lesion (structural lesion in the brain). Apasmara is the spectrum of diseases where no organic lesion is observed after elaborate evaluation of the patient.
- Apatanaka and Apatantraka are categorized under Hridroga as per Caraka.
- Dandapatanaka or Dandaka is attributable to the state of status epilepticus.
- Ardita is the condition where the person will have manifestation of the disease in term of Vega (episodes). Pakshaghata is the spectrum of the diseases, where clinical manifestations are observed continuously without any symptom free period.
- Antarayama and Bahirayama are the clinical manifestations where bending of the body parts are seen. These can be attributable to decorticate and decerebrate lesions respectively.
- Gridhrasi is the disease where Vata Dosha vitiation is observed, where the person may have appearance of the pain in the buttock region, radiates to whole lower limb.
- Vishwachi is the spectrum of the disease, where Vata Dosha vitiation is being noted, where the clinical presentation is restricted to upper limb.

Self-Assessment: Questions for Reviews

Long essay questions

- Explain Ardita
- Describe Pakshaghata
- Explain Antarayama and Bahirayama

Short essay questions

- Explain Akshepaka
- Describe Apatantraka
- Explain Gridhrasi
- Write the differences between Khanja and Gridhrasi

MCQs

1. Vata and Vatakaphaja varieties of the disease are found in

a. Vishwachi; b. Gridhrasi;
c. Pakshaghata; d. Ardita

Answer: (b)

2. In Khanja, which of the following is involved?

a. One upper limb;
b. One lower limb;
c. Both upper limbs;
d. Both lower limbs

Answer: (b)

3. Which of the following is appropriate answer in the patient of Vishwachi?

a. The patient will have radiation of the pain to the any one of the lower limb;
b. Pain radiates to any of the upper limb;
c. Pain radiates in both upper and lower limbs;
d. Pain radiates to one side of upper and lower limb

Answer: (b)

4. The type of examination done to assess consciousness of the patient is

a. EMV score;
b. Apgar score;
c. MRC score of motor system;
d. Sensory system examination

Answer: (a)

5. Which of the following disease is having the poor prognosis of the disease after 3 years of duration of the illness?

a. Ardita; b. Pakshaghata;
c. Akshepaka; d. Apatanaka

Answer: (a)

Chapter resources

- Charaka Samhita Chikitsa Sthana – 28th chapter: Vatavyadhi Chikitsa Adhyaya
- Sushruta Samhita Nidana Sthana – 1st chapter: Vatavyadhi Nidana Adhyaya
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